

Brain-computer UX is a trust-and-measurement problem

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CITATION GATE

Cleared 2026-09-04. Every item previously graded **LIKELY** was re-verified against a primary source and upgraded, including both neural-privacy bill numbers, confirmed against the legislatures' own texts. One company, Subsense Inc, could not be verified and is marked as such rather than described.

ABSTRACT

Neural-interface research optimizes the decoder. It has spent two decades improving how accurately a machine reads a brain, measured almost entirely by one metric — Information Transfer Rate. But ITR measures decoder throughput, not whether a system is usable, trustworthy, or sustainable to live with. This paper argues that the *other half* of the neural interface — how it presents its read of you back to you, how much it costs you to use, and whether you can calibrate trust in a system that infers your own internal state — is under-instrumented and under-theorized, and that it is a **measurement-and-trust problem a behavioral scientist is positioned to own, not a hardware problem**. The claim is deliberately narrow: the field is not empty — passive-BCI and neuroergonomics researchers have touched every piece of this — but no one has approached it as a first-class measurement-and-trust-calibration problem the way a behavioral scientist would. Six concrete, instrumentable open problems follow.

1. The metric only measures one side

WHAT ITR COUNTS

number of classes

classification accuracy

trial duration

WHAT IT HAS NO TERM FOR

user effort

fatigue over a session

dropout

whether the person trusts the read

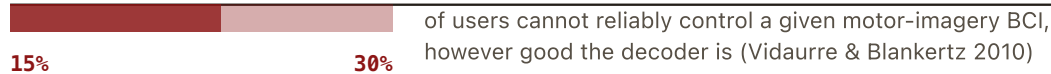


FIGURE 1 – THE METRIC AND ITS BLIND SIDE

Information Transfer Rate is built from three decoder terms and carries no term for anything the user experiences. A system can post a high number and be exhausting inside ten minutes. The band underneath is the share of people who cannot drive a motor-imagery BCI at all regardless of decoder quality, and part of that turns out to be an interface artifact rather than a biological ceiling.

The dominant performance metric is Information Transfer Rate (Wolpaw et al. 2002; bits/min from class count, accuracy, trial duration) — VERIFIED. It rewards more classes and faster trials, with **no term for user effort, fatigue, dropout, or subjective experience**. A system can post a high ITR and be exhausting to use for ten minutes.

Alongside it sits a problem hardware has not solved: **BCI "illiteracy."** Roughly 15–30% of users cannot reliably control a given motor-imagery BCI regardless of decoder quality (Vidaurre & Blankertz 2010 — VERIFIED; Allison & Neuper 2010 — VERIFIED — frame it as a population-level *design* failure, not a user failure). Crucially, part of it is an interface artifact, not a biological ceiling: Carabalona (2017, *Front. Neurosci.* — VERIFIED) shows P300 "illiteracy" is partly a stimulus-design problem. And session performance degrades as attention and fatigue degrade the very signal the system depends on — a vicious loop (Myrden & Chau 2015 — VERIFIED), instrumented from psychology (NASA-TLX) but never built into the interface itself.

The gap, stated as a measurement problem: the field has a metric for how many bits the decoder extracts and no adopted metric for what those bits cost the user. A validated "usable bits/min" or session-cost index — effort, fatigue, dropout risk, subjective burden, comparable across studies — is a concrete, citable instrument, exactly the specialty a measurement-builder brings.

2. Passive BCI inverts the human-factors problem

The product-relevant branch is passive / neuroadaptive BCI (Zander & Kothe 2011, the active/reactive/passive taxonomy — [VERIFIED](#)). Here the user does not intentionally drive anything; the system reads state (workload, attention, engagement, error-awareness) and acts. The mature real-world use is safety-critical operator monitoring: adaptive automation for air-traffic controllers (Di Flumeri et al. 2019 — [VERIFIED](#)), dry-EEG for drivers (Zander et al. 2017 — [VERIFIED](#)), adaptive instruction under cognitive load (Gerjets et al. 2014; Beauchemin et al. 2024 — [VERIFIED](#)), and the case for mental-state-based adaptive interfaces over long sessions (Hinss et al. 2022 — [VERIFIED](#)). This is the branch closest to funded, defensible deployment — aviation, defense, industrial — as opposed to earlier-stage consumer neurotech, and it is the branch that ties directly to human-machine teaming.

The inversion: passive systems don't need to solve intentional-control illiteracy, but they inherit a harder problem. **State inference is probabilistic and often wrong, and the system now acts on a guess about your internal state.** The human-factors question shifts from "can I drive this" to "do I trust what it just did, and did it do the right thing."

3. The readback problem: no framework for what a neural interface should show you

There is an active neurofeedback-modality literature, but it answers "which sensory channel best *trains* a target brain state," not "what information architecture, timing, and framing should govern how a system communicates its inferred read of you *back* to you." Those are different questions, and the second — the UX/trust question — is essentially unframed. Reviews categorize feedback mechanisms (Zhang et al. 2026 — [VERIFIED](#)) without prescribing which to choose for a given cognitive-load or attention state; the field's own researchers gesture at the gap ("more entertaining, multimodal, adaptive feedback might improve UX" — Kober, Wood & Berger 2024, [VERIFIED](#)) without formalizing it.

Neural interfaces carry a wrinkle no adjacent field has: **feedback about your own inferred state can change that state** — telling someone "you look distracted" measurably shifts attention and arousal (an observer effect unique to this domain). Almost no reviewed work treats this as a design constraint. Aviation alarm-design (urgency × modality × false-alarm cost) and mobile-HCI interruption research (interruption cost as a function of task state) are the analogues that exist elsewhere and do not exist for neural readback. **Status:** [OPEN](#). The honest claim is not "nobody has looked" but "the pieces are touched, adjacent to their authors' specialties; no transferable design framework exists."

4. The trust problem is a cognitive-attack-surface problem pointed inward

Automation-bias research shows people mis-calibrate trust in automated agents — too much or too little (de Visser et al. 2018 — [VERIFIED](#)). But that research was built for systems acting on *external* data the user can independently check: radar, a thermostat. **Neural state inference removes the independent check.** When a neuroadaptive system reallocates a task, suppresses a notification, or escalates an alert

because it inferred you were overloaded, you often cannot verify or contest its read of *you*. That is a categorically different trust problem than trusting a system's read of the external world, and I did not find literature that names the asymmetry and treats it as the design problem it is ([OPEN](#)).

This is the same structure as the Cognitive Attack Surface, turned inward: a system acting on a probabilistic model of your internal state, with the disclosure and calibration questions front and center. The countermeasures transfer — confidence display, disclosure of the inferred read, override affordances that don't themselves spike cognitive load, calibration instrumentation — and designing them is a measurement-and-trust problem, not a decoder problem. Ehrlich et al. (2023, [VERIFIED](#)) already show systems acting *preemptively* on anticipatory neural responses, which is where "who decided that" stops being philosophical.

5. The governance context is arriving fast

Neurorights is now a live legislative area, which makes trust and disclosure not just UX but compliance: Canada federally protected neural data (Feb 2026 — [VERIFIED](#) via the Neurorights Foundation); the US MIND Act was introduced Sept 2025 ([VERIFIED](#)); Colorado (HB24-1058, signed 2024-04-17, in force 2024-08-07) and California (SB 1223, signed 2024-09-28, Chapter 887, in force 2025-01-01) amended their privacy acts to classify neural data as sensitive ([VERIFIED](#) — both bill numbers confirmed against the legislatures' own texts). A system that infers internal state and cannot explain or let the user contest its read is walking into this. Trust-calibration design is becoming a regulatory surface, not only a UX nicety.

6. Six open problems a measurement specialist can own

Each is a problem a behavioral scientist — not a hardware or ML engineer — is the right kind of expert for:

1. **A usability metric beside ITR** — a validated "usable bits/min" / session-cost index (effort, fatigue, dropout, subjective burden), comparable across studies. A literal measurement-instrument contribution.
2. **A feedback-timing/modality framework for neural readback**, grounded in attention and cognitive-load theory rather than case-by-case modality trials — designed experiments manipulating interruption cost, framing, and modality against measured attention/arousal.
3. **Trust-calibration instrumentation for neuroadaptive systems** — measuring, in real time, whether a user over- or under-trusts the system's read of their own state, and designing corrections (disclosure, confidence, override) that don't spike load. The cognitive-security toolkit, pointed inward.
4. **Illiteracy as a design variable, not a user trait** — treating "can't produce a usable signal" as an interaction-design failure to test and iterate, following Carabalona's crack.

5. **Platform-scale interaction standards** for when BCI becomes an OS input method (Synchron × Apple is the leading indicator, May–Aug 2025 — VERIFIED) — the conventions touch and voice each needed, built by HCI researchers, not hardware engineers, and not yet built for neural input.

6. **Reward/engagement design for neurofeedback and closed-loop training** — variable-reward and engagement-without-surveillance-anxiety as first-class design variables, which the neurofeedback literature treats as a training-efficacy afterthought. Directly the reward-circuitry work of the Cognitive Attack Surface papers.

7. The honest thesis

Not "nobody has looked at neural-interface UX" — Zander/Kothe-lineage passive-BCI work, Hinss et al. (2022), and Gerjets et al. (2014) each touch pieces of it, and a credible paper cites them as partial precedent. The sharper, defensible claim: **the pieces have been touched by neuroscientists and HCI researchers working adjacent to their own specialty; no one has approached neural-interface UX as a first-class measurement-and-trust-calibration problem the way a behavioral scientist would.** That is the seat this program takes.

Note on the industry frame

Company facts (Neuralink human trials from Jan 2024; Synchron's Apple/NVIDIA OS-integration play; Precision Neuroscience's April 2025 510(k); Blackrock's company-reported metrics — treat as company claims) were live-verified this pass and date the inflection toward platform-scale neural UX. **Subsense Inc could not be verified** — no public footprint was found this pass (subsense.ai is a parked domain). It is likely an early/stealth company; confirm what it builds directly with the contact before citing it. Do not describe Subsense's product from assumption.

References

All entries verified against a primary source or the issuing body on 2026-09-04.

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Legislation. Colorado HB24-1058 (2024), amending the Colorado Privacy Act to define biological and neural data as sensitive data; signed 2024-04-17, in force 2024-08-07. California SB 1223 (2024), Chapter 887, amending the CCPA to classify neural data as sensitive personal information; signed 2024-09-28, in force 2025-01-01. Canada federal neural-data protection (Feb 2026) and the US MIND Act (introduced Sept 2025) per the Neurorights Foundation.

Unverifiable, and marked as such. Subsense Inc could not be verified: no public footprint was found (subsense.ai is a parked domain). It is likely early or stealth. Do not describe its product from assumption; confirm directly with the contact before citing.

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Every citation graded on the page. Sources that could not be verified are printed as unverified.